IS460 MACHINE LEARNING · PROJECT PROPOSAL

Predicting Airline Passenger Satisfaction via Deep Tabular Ensembles: Evaluating Synthetic Fidelity and Explainable Drivers



GROUP 03 DATA 829,515 PASSENGERS METRIC ROC AUC MODELS 3 FAMILIES

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Problem Statement

A post-flight survey tells an airline who was unhappy after the fact. We predict each passenger's probability of being satisfied, and explain which services drive it.

INPUT	21 features [1]: 4 traveller, 4 trip and 13 service-rating fields
OUTPUT	P(satisfied) for each passenger, plus the services that move it
METRIC	ROC AUC: the chance a satisfied passenger is ranked above an unsatisfied one [2]
DATA	699,635 synthetic passengers, generated from 129,880 real survey responses [1]

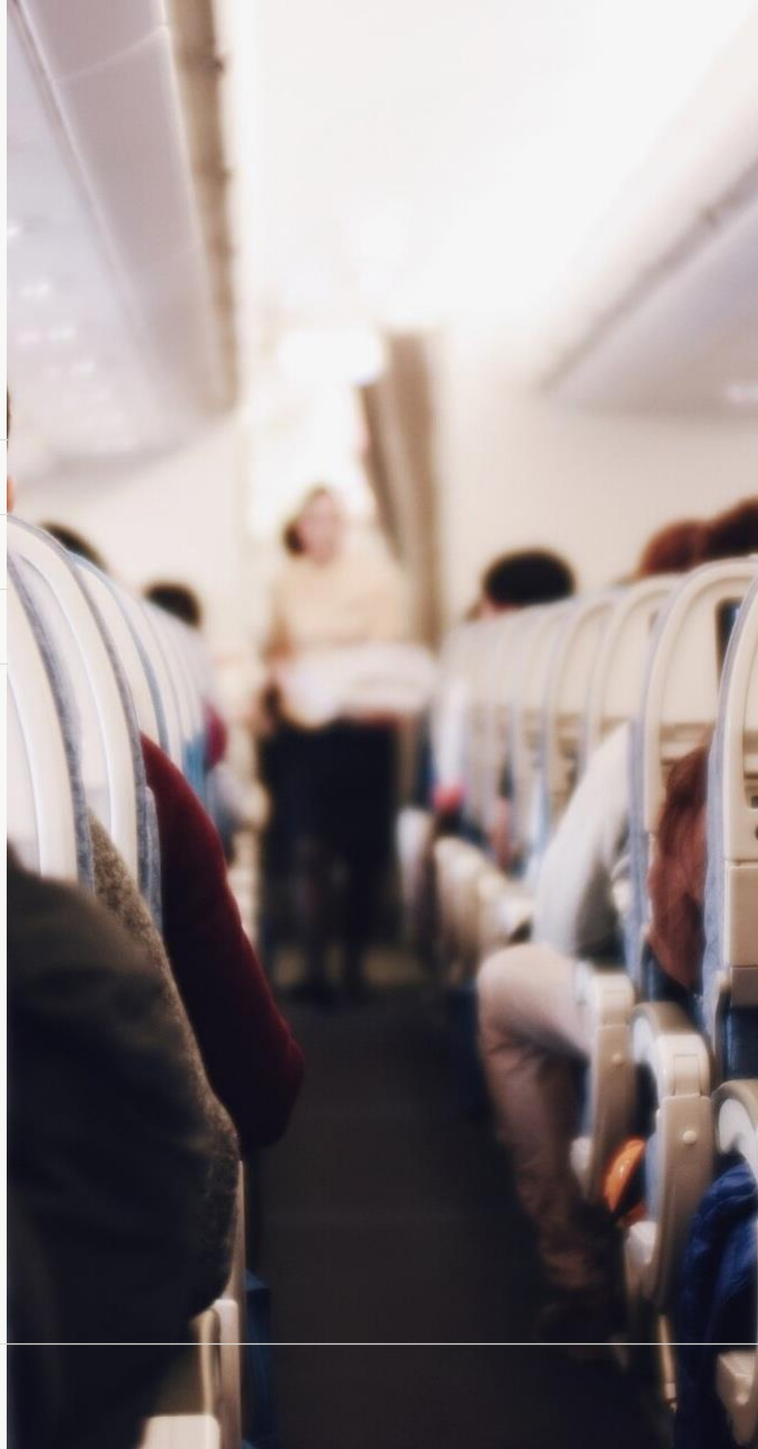


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[1] Y. Yan, W. Reade, and E. Park, "Predicting Airline Satisfaction," Kaggle Playground Series S6E10, 2026.

[2] J. A. Hanley and B. J. McNeil, "The meaning and use of the area under a receiver operating characteristic (ROC) curve," *Radiology*, vol. 143, no. 1, 1982.

Motivation

5.1 bn
air passengers expected in 2026, with a record 84% of seats filled [3]

US$4.5
forecast net profit per passenger in 2026, down from US$9.1 [4]

76/100
US airline satisfaction in 2026; in-flight internet ratings up 20% [5]

5–25×
more costly to win a new customer than to keep one [6]

Satisfied passengers fly again and recommend the airline [7]. On margins this thin, knowing **which service to fix first** is worth real money.

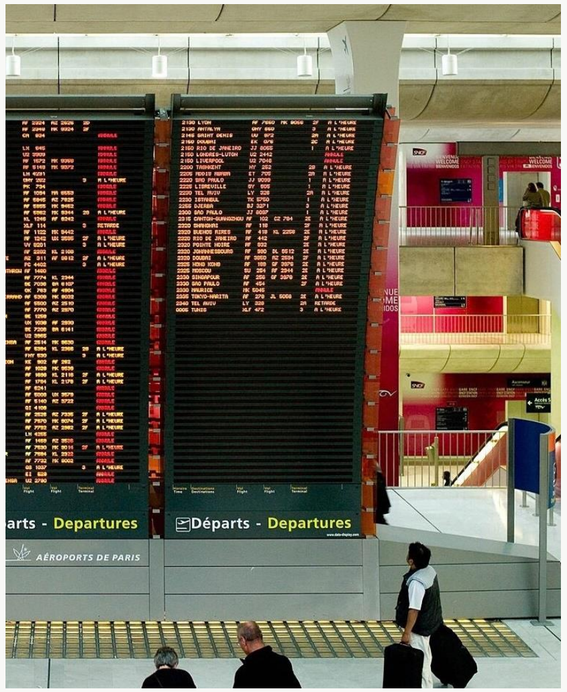


Photo: Joi Ito, CC BY 2.0, cropped

[3] IATA, "Middle East disruptions and high fuel prices halve airline industry profitability," press release, Jun. 7, 2026.
[4] IATA, "Industry statistics fact sheet," Jun. 2026.
[5] ACSI, "ACSI Travel Study 2026," press release, Apr. 21, 2026.
[6] A. Gallo, "The value of keeping the right customers," *Harvard Business Review*, Oct. 29, 2014.
[7] M. Kos Koklic, M. Kukar-Kinney, and S. Vegelj, "An investigation of customer satisfaction with low-cost and full-service airline companies," *J. Bus. Res.*, vol. 80, 2017.

LITERATURE REVIEW · 1 OF 4

Service Quality and Satisfaction

1988

Parasuraman et al.

Satisfaction tracks the gap between expected and perceived service [8].

2004

Park et al.

Service value and satisfaction drive passengers' future intentions [9].

2008

Chen

Perceived value builds satisfaction, which drives loyalty [10].

2009

An and Noh

What drives satisfaction differs by cabin class [11].

2020

Bellizzi et al.

Review of airline service-quality survey studies [12].

2022

Noviantoro and Huang

Data mining ranks online boarding and Wi-Fi as top levers [13].

[8] A. Parasuraman, V. A. Zeithaml, and L. L. Berry, "SERVQUAL: A multiple-item scale for measuring consumer perceptions of service quality," *J. Retailing*, vol. 64, no. 1, 1988.

[9] J.-W. Park, R. Robertson, and C.-L. Wu, "The effect of airline service quality on passengers' behavioural intentions: A Korean case study," *J. Air Transp. Manag.*, vol. 10, no. 6, 2004.

[10] C.-F. Chen, "Investigating structural relationships between service quality, perceived value, satisfaction, and behavioral intentions for air passengers: Evidence from Taiwan," *Transp. Res. A*, vol. 42, 2008.

[11] M. An and Y. Noh, "Airline customer satisfaction and loyalty: Impact of in-flight service quality," *Service Business*, vol. 3, no. 3, 2009.

[12] M. G. Bellizzi, L. Eboli, and G. Mazzulla, "Air transport service quality factors: A systematic literature review," *Transp. Res. Procedia*, vol. 45, 2020.

[13] T. Noviantoro and J.-P. Huang, "Investigating airline passenger satisfaction: Data mining method," *Res. Transp. Bus. Manag.*, vol. 43, 2022.

Machine Learning on This Survey

STUDY	MODELS	BEST REPORTED
Jiang et al., 2022 [14]	Random forest + feature elimination, LR, KNN, NB	RF: accuracy 0.963
Zhang, 2024 [15]	LR, random forest, AdaBoost, XGBoost + SHAP	XGBoost: AUC 0.987
Lee, 2024 [16]	Bagging, boosting and stacking ensembles	Stacking: accuracy 0.963
Bilgin, 2025 [17]	Gradient-boosted trees vs deep nets (DNN, GRU)	LightGBM and GRU best
Hyassat et al., 2026 [18]	Gradient-boosted trees with SHAP importance	LightGBM: AUC 0.994

Takeaway: tuned boosted trees reach AUC 0.98–0.99; SHAP studies agree Wi-Fi and online boarding are leading predictors [15], [18].

[14] X. Jiang, Y. Zhang, Y. Li, and B. Zhang, "Forecast and analysis of aircraft passenger satisfaction based on RF-RFE-LR model," *Sci. Rep.*, vol. 12, 2022.

[15] Y. Zhang, "Satisfaction analysis of airline passenger experience," in *Proc. EMITI*, 2024, pp. 141–152.

[16] M. Lee, "Comparison of bagging, boosting, and stacking ensemble models for airline customer satisfaction analysis," *FaST*, vol. 8, no. 1, 2024.

[17] S. N. Bilgin, "Prediction of airline passenger satisfaction using machine learning and deep learning approaches," in *Proc. IEEE ASYU*, 2025.

[18] A. S. Hyassat, M. O. Hiari, and M. M. Abualhaj, "From data to strategy: Explainable AI insights for enhancing airline passenger satisfaction and loyalty," *J. Artif. Intell. Technol.*, vol. 6, 2026.